

Part 1: Introduction and Data Exploration

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1. Research Questions and Null Hypotheses

The objective of this project is to identify and evaluate the key determinants of housing prices in Ames, Iowa. The null hypotheses examined in this study are as follows:

- The overall quality of a house has no relationship with its sale price.
- Newer homes and older homes have the same average sale price.
- Homes located in different neighborhoods have the same average sale price.
- Homes with basements and homes without basements sell for the same price.
- Homes with central air conditioning sell for the same price as homes without central air conditioning.
- The amount of above-ground living space has no effect on the sale price of a home.
- Homes with different levels of functional condition have the same average sale price.
- The quality of the kitchen has no relationship with the sale price of a home.
- The presence or quality of a swimming pool has no relationship with a home's sale price.
- The number of full bathrooms above ground is not associated with a home's sale price.
- The total number of bathrooms in a home is not associated with its sale price.
- The number of bedrooms above ground is not associated with a home's sale price.
- The total number of rooms above ground is not associated with a home's sale price.
- The quality of the fireplace has no relationship with a home's sale price.
- The size or capacity of the garage has no effect on a home's sale price.

2. Dataset Description

This project uses the Ames Housing Dataset as the primary research dataset. Originally constructed by Dean De Cock, the dataset contains 2,930 residential property transactions from the city of Ames, Iowa. Each observation provides 79 features describing structural attributes, location information, construction and remodeling history, quality assessments, and various home amenities. These features span numerical, categorical, and ordinal data types. Owing to its rich information content and diverse variable structure, the Ames Housing Dataset has become a widely used benchmark for developing and testing machine learning algorithms and techniques.

3. Review of Prior Work

Many previous analyses on Kaggle for the housing dataset have used linear or regularized regression models to understand which home characteristics drive sale prices. Common findings across these projects are that overall material and finish quality (OverallQual) and above-ground living area (GrLivArea) are strong predictors of price, often explaining a large portion of the variation in log-transformed sale prices. Neighborhood indicators, year built or remodeled, and garage size are also repeatedly identified as important.

A broader review by Sirmans, Macpherson, and Zietz (2005)^[1] supports these patterns across 125 real-estate studies. Their analysis found that square footage, number of rooms, lot size, overall quality, age, and neighborhood location are consistently significant predictors of sale price. Because these conclusions were based on linear regression frameworks similar to those used in this project, they reinforce the importance of size, quality, age, and location when modeling housing prices.

4. Exploratory Analysis

To explore which numeric variables are strongly associated with sale price, we computed Pearson correlations between SalePrice and all numeric predictors in the dataset. The variables with the highest correlations were OverallQual, GrLivArea, GarageCars, GarageArea, and TotalBsmtSF. Year-related features such as YearBuilt and YearRemodAdd also showed moderate positive correlations. Only a small number of predictors showed near-zero or slightly negative correlations, suggesting weak or negligible linear relationships with sale price. Overall, the correlation structure highlights size, quality, and age as the most influential predictors for our regression analysis.

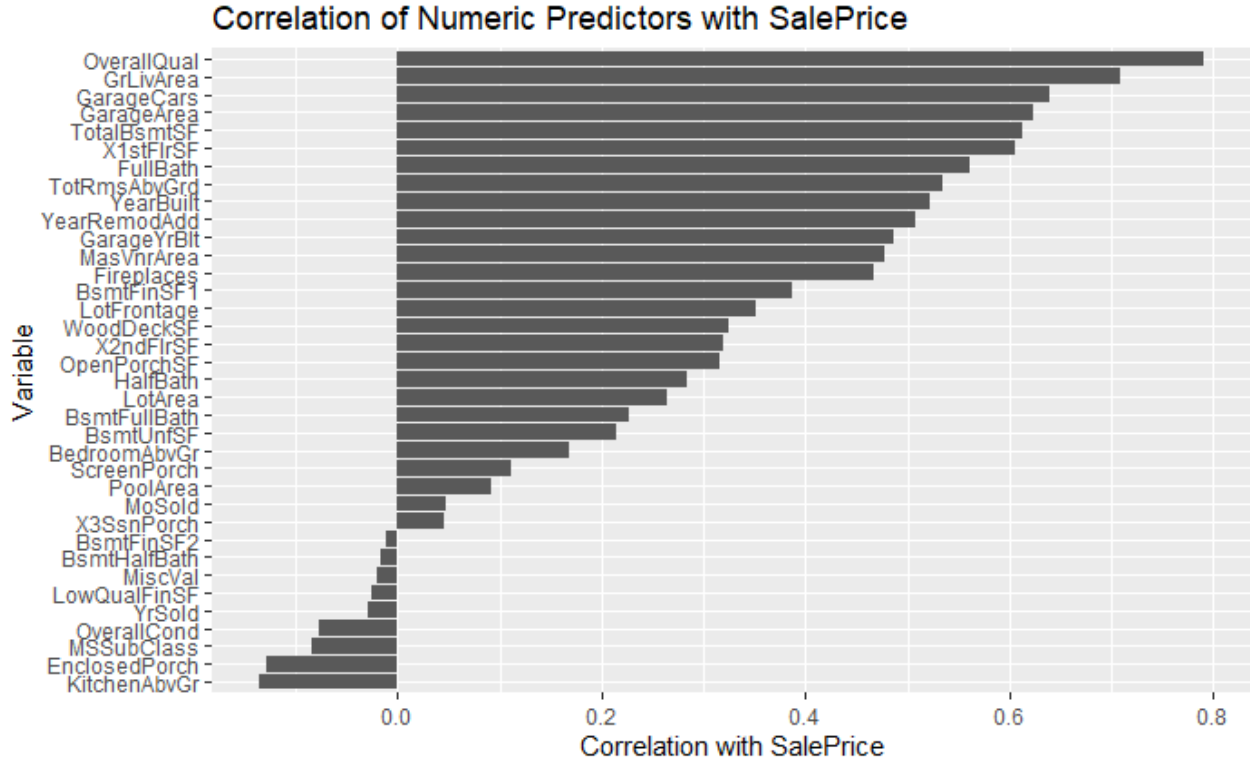


Figure 1: Correlation Plot

5. Data Quality Assessment

An initial review indicates several data considerations relevant to analysis. Only a few predictors contain genuine missing values, as most NA entries reflect the true absence of a feature. LotFrontage accounts for the majority of missingness (17.7%), while MasVnrArea and Electrical contain only eight and one missing values. The dataset also includes a small number of notable outliers, such as homes with unusually large living areas paired with unusually low sale prices.

Several numerical variables show pronounced right skewness, including PoolArea and LotArea, and SalePrice shows a similar pattern. Strong correlations among major size-related predictors highlight the possibility of multicollinearity. Despite these considerations, the Ames Housing Dataset remains highly suitable for addressing the research objectives. Its large sample size and broad set of well-documented features support rigorous analysis of the determinants of housing prices.

6. Analysis Plan

For the formal analysis in Part 2, we will fit several models to study the determinants of housing prices and to predict sale prices accurately. The primary outcome will be `SalePrice`, though we will also model $\log(\text{SalePrice})$ to address skewness and improve linearity. Predictors will include variables describing home size, quality, age, and location, with categorical variables converted into one-hot encoded indicators.

In addition to ordinary least squares (OLS) regression, we will fit regularized and nonlinear models such as lasso regression and random forests to capture interactions and nonlinear effects. For linear models, inferential objectives include estimating effects, constructing confidence intervals, and testing hypotheses for individual and grouped predictors. For flexible models, emphasis shifts toward prediction accuracy and variable importance.

Model adequacy for linear regression will be evaluated using standard diagnostic plots and influential point analysis. To compare predictive performance across models, we compute R^2 , adjusted R^2 , RMSE, and MSE on validation and test datasets, with random forests serving as the primary non-linear benchmark. This evaluation strategy allows us to identify influential predictors while determining whether flexible models provide improvements in out-of-sample accuracy.

Acknowledgements

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Reference

[1] Sirmans, G. Stacy, David A. Macpherson, and Emily N. Zietz. “The Composition of Hedonic Pricing Models.” *Journal of Real Estate Literature*, 13(1), 2005, pp. 1–44.